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MIDDLE EAST TECHNICAL UNIVERSITY

NORTHERN CYPRUS CAMPUS

ASE 446: HELICOPTER AERODYNAMICS AND DESIGN

FINAL PROJECT REPORT

SAR HELICOPTER

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Introduction

The aim of this report is to discuss the design of a double engine, highly maneuverable, medium weight, Search and Rescue Helicopter capable of lifting medium to high weight loads for extended ranges and a large endurance value. The design characteristics are accommodated to suit the region of operation and the general demand of both civil and private needs. Some physical parameters have been assumed and designed around competitor Search and Rescue Helicopters already established in the current world.

Our design characteristics have been based off the operating regions of Oslo Norway. The city houses an expansive mountainous region as well as large bodies of water, that require long range and maritime operations. Oslo being the capital and having a large population with a booming tourist industry, the demand for several SAR helicopters is already in the market.

The ideal scenario for any SAR is to have its production come under a cheap cost without compromising its ability for passenger safety or rescue. Hence it is absolutely necessary for our design to meet the competitive performance of existing modals that can meet the criteria required. We used both theoretical/industry methods to make reasonable weight and power estimations. And using extensive literature research selected the most optimal design structures for needs.

Chapter 1. Design Requirements

The minimum design requirements for our SAR helicopter need to meet a standard that exceeds most SAR helicopters as depicted below.

1.1 Minimum Requirements

- Cruise speed: > 230 km/h
- Range (no reserve at cruise speed): > 400 km
- Endurance: 5 hours
- Service ceiling: 4,200 m
- Useful payload: 2 pilots + 10 passengers + special equipment
- Empty weight fraction: < 0.45
- Noise: External (driven by tip speed)
    - Main rotor tip speed: < 195 m/s
    - Tail rotor tip speed: < 180 m/s
- Cooper Harper rating of 3.5 or less (Level I Handling Qualities)

1.2 Additional Design Criteria

- Configuration: Conventional
- Engine: 2 x Turboshaft
- Landing Gear: Skid Type
- City of Operation: Oslo
- Main Rotor Configuration: 4 bladed hinge-less semi rigid
- Empennage: Conventional
- Tail rotor: 2 Bladed
- Spacious
- Easy access
- Stable and Safe

Chapter 2. Competitor Study

Taking our requirements and standards in mind, we conducted a competitor study through extensive research into preexisting Search and Rescue helicopters. Their general characteristics and performance parameters are shown below in their respective tables.

2.1 General Characteristics

Parameter	Leonardo AW 139	Bell 212	Sikorsky S-92	Sikorsky HH-60	Sikorsky 70I
Crew + Passenger capacity	2+15	2+14	2+19	2+12	4+11
Length	54 ft 8 In	57 ft 1.7 In	68 ft 6 In	64 ft 10 In	64.83 ft
Width	10 ft	8 ft 7 In	17 ft 3 In	18 ft 1 In	14.34 ft
Height	16 ft 4 In	12 ft 6.8 In	15 ft 5 In	16 ft 8 In	17.49 ft
Main Rotor Diameter (Feet inch)	45 ft 3 In	48 ft	56 ft	52 ft 8 In	53.68 ft
Number of Blades	5	2	4	4	4
Empty Weight (lbs)	7985	6529	15500	16000	11790
Maximum Gross Weight (lbs)	14110	11200	27700	22000	18000
Power Plant	2 Turboshaft	1 Turboshaft	2 Turboshaft	2 Turboshaft	2 Turboshaft
Horsepower (Hp)	1531	1743	2520	1940	1944

Fuel Capacity (Liters)	1568	821	2877	2565	1360
Unit Cost (USD)	11M	4.9M	27M	15.8M	13M

Table 1. General Characteristics

2.2 Performance Parameters

Parameter	Leonardo AW 139	Bell 212	Sikorsky S-92	Sikorsky HH-60	Sikorsky 70I
Cruise Speed (Mph)	191	120	174	184	180
Max Speed (Mph)	193	140	190	224	224
Range (Miles)	659	273	620	373	290
Service Ceiling (ft)	20000	17400	14000	14000	15000
Rate of Climb (ft/min)	2140	1745	1600	1765	2250
Max Operating altitude (ft)	15000	20000	15000	18000	20000
Disc Loading (lbs/ft^2)	9.22	6.2	9.8	9.06	7.95

Table 2. Performance Characteristics

Chapter 3. Sizing and Performance Analysis

3.1 Engine Selection

The Pratt & Whitney Canada PT6C-67C engine is our ideal design choice due to its balance of power and reliability. We use two engines in our design. With a power output of 1531 horsepower per engine, it provides

- Higher performance and greater payload capacity.
- Making it suitable for missions search and rescue.
- Offshore transportation.
- Other mission-critical operations.

Furthermore, the PT6 engine maintenance costs are low and designed to operate in extreme conditions, such as high altitudes and hot temperatures, making it suitable for search and rescue missions in Oslo, Norway.

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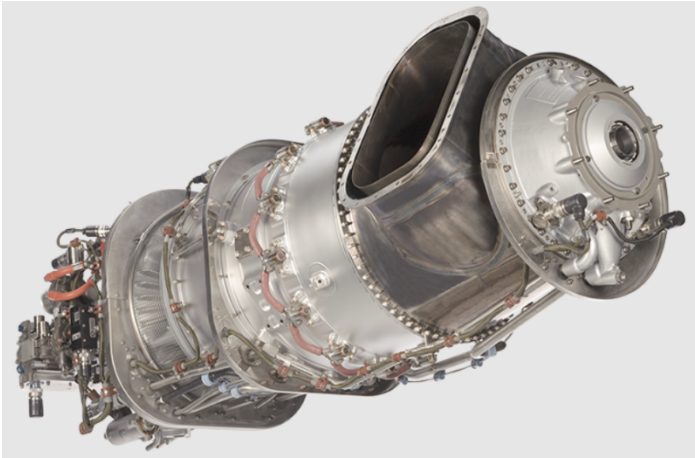


Figure 1 PT6C-67C

3.2 Gross Weight Estimation

In this subchapter, we discuss the methodology and results of the estimation of the gross take off weight of our helicopter design. Using statistical modal equations for our weight groups and summing those groups together. We firstly use three different values of Disk Loading selected from the competitor study. And using that to calculate the radius of the main rotor and furthermore the thrust and power required to lift those weight groups under hover.

It is important to note that throughout our calculations for sizing we made two very important and distinct assumptions

- The helicopter operates in hover throughout, hence making our thrust be equal to Weight
- And that all Main Rotor and Tail Rotor blades are rectangular in shape and zero taper.

To initialize our calculations, we must first make an initial guess weight by comparing typical values of our competitors and hence we selected the initial takeoff weight as

$$W_G = 1100\ lb.$$

From competitor study, we see that disk loading values range from 2.4-3.4 lb/ft^2. The value of disk loading designs for our helicopter lies within this range and hence use that value to estimate its true total gross weight. The weight estimation is done using the following values:

$$DL_1 = 9.5\frac{lb}{ft^2}\ ,\ \ DL_2 = 10\frac{lb}{ft^2}\ ,\ DL_3 = 14\frac{lb}{ft^2}$$

The formula for calculating the main rotor radius is given as

$$R_{MR} = \sqrt{\frac{w_G}{\pi \times DL}}$$

From literature, tail rotor radius is estimated to be about a fifth of the main rotor radius. This gives

$$R_{TR} = 0.2R_{MR}.$$

The solidity for the main rotor is the ratio of blade area to disk area

$$\sigma_{MR} = \frac{A_b}{A} = \frac{N_b \times c}{\pi \times R_{MR}},$$

where σ_{MR} is the solidity of the main rotor, N_b is the number of blades in the main rotor, c is the cord length of the blade and R_{MR} is the radius of the main rotor. Literature tells us that solidity values ranging from 0.06 to 0.12 are ideal and so a value of 0.10 was used.

The tail rotor solidity is assumed to be in the same magnitude with the main rotor solidity as observed from historical trends. Hence

$$\sigma_{MR} = \sigma_{TR}.$$

Part 2: Weight groups

1. Main rotor group (W_R)

$$W_R = 1.7 \times {W_G}^{0.342} \times {R_{MR}}^{1.58} \times {\sigma_{MR}}^{0.63}$$

where W_R is the weight of the main rotor group and W_G is the take-off gross weight of our helicopter.

2. **Tail rotor group (W_{TR})**

$$W_{TR} = 7.12 \times \left(\frac{W_G}{1000}\right)^{0.446} \times {R_{TR}}^{1.62} \times {\sigma_{TR}}^{0.66}$$

where W_{TR} is the weight of the tail rotor group and R_{TR} is the radius of the tail rotor

3. **Power plant group (W_T)**

$$W_T = 0.166 \times HP$$

where W_T is the weight power plant group and HP is power available of our selected engine. This is the power output of the Konner TK-250 turboshaft engine.

4. **Power plant section group (W_{PS})**

$$W_{PS} = 0.00155 \times {W_G}^{1.07} \times DL^{0.54},$$

where W_{PS} is the weight of the power plant section group in and DL is disc loading.

5. **Mechanical drive system group (W_{DS})**

$$W_{DS} = 42.4 \times \left(\frac{HP \times R_{MR}}{V_T}\right)^{0.763}$$

Where W_{DS} is the weight of the mechanical drive system and V_T is the main rotor tip speed

6. **Flight control group (W_{FC})**

$$W_{FC} = 0.0226 \times {W_G}^{0.712} \times {V_C}^{0.653}$$

where W_{FC} is the weight of the flight is control group and V_C is the cruise velocity of the design helicopter. From the competitor study, V_C varies between 60 knots and 100 knots therefore V_C is chosen as 85 knots to have a fast and mobile helicopter for our desired range.

7. **Landing gear group (W_{LG})**

$$W_{LG} = 0.475 \times {W_G}^{0.975}$$

Where W_{LG} is the weight of the landing gear group.

8. **Fuselage group (W_F)**

$$W_F = 0.0382 \times {W_G}^{0.998} \times L^{0.942} \times W^{0.453} \times H^{0.295}$$

9. Forward propulsion group (Wp)

$$W_P = 0.146 \times \left(\frac{W_G}{Ip}\right)$$

where W_F is the weight of the fuselage group, L is the length of the fuselage, W is the width of the fuselage and H is the height of the fuselage. The values of L, W, and H are obtained as 7ft, 5.5 ft, and 7.8 ft respectively from competitor study. For a small utility, the above formula simplifies to the following formula which is used in the MATLAB.

$$W_F = 0.21 \times {W_G}^{0.598} \times {R_{MR}}^{0.942}$$

Part 3: Crew and payload weight

We have 2 pilots and 12 passengers. Each person is assumed to weigh a maximum of 80 kg and a specific baggage allowance. Therefore, crew and payload weight are obtained as

$$W_{crew} = 195\ lb,$$

$$W_{payload} = 176.37 + 33.0693 = 209.439\ lb,$$

where two pilots with 195 lbs including luggage (80 kg + 8kg) and one passenger with 210 lb with luggage (80 kg + 15 kg luggage).

Part 4: Fuel weight

Helicopters which cover a range almost like our design and carry a similar payload are analyzed for the competitor study. The fuel tank capacity is taken from competitor study as 140 liters. The fuel density is taken from JET A1 fuel at 15℃ temperature as 0.804 kg/m^3. Therefore, the fuel weight is as follows

$$W_{fuel} = \rho \times V = 250 \text{ lb.}$$

Part 5: Gross weight estimation [7]

The empty weight of the design helicopter is estimated by summing up the weight groups. Therefore, adding all these weights gives the empty weight,

$$W_{empty} = W_R + W_{TR} + W_T + W_{PS} + W_{DS} + W_{FC} + W_{LG} + W_F + W_W + W_p.$$

Finally, the gross take-off weight of the design helicopter is obtained as

$$W_G = W_{empty} + W_{crew} + W_{payload} + W_{fuel}.$$

Summary of the results from MATLAB code

Disk Loading	DL=9.5	DL=14	DL=12
Main Rotor radius(ft)	18	21.6	20
Gross weight (lb)	12355	12480	12370
Empty weight Fraction	0.4486	0.4541	0.4518

Table 3 Weight and Radius Estimates

We chose DL=9.5 since it gives the smallest Gross weight.

Weight Group	Weight Result (lb)
Main Rotor Group	241.25
Tail Rotor Group	26.96
Power Plant Group	251.97
Power Plant Section Group	124.90
Mechnical Drive System Group	5543.3
Flight Control Group	636.02
Landing Gear Group	463.7
Fuselage Group	2251.3
Forward Propulsion Group	220.23
Fuel Weight	2765
Empty Weight	5543
Gross weight	12355

Table 4 Design Weight Groups

3.3 POWER REQUIREMENTS

In this section, the power required in forward speed is calculated. This is done using Blade Element Theory as show below

$$P_T = P_{Induced} + P_{Profile} + P_{Parasite},$$

where P_T is the total power required, $P_{Induced}$ is the induced power, $P_{Profile}$is the profile power and $P_{Parasite}$ is the parasite power during forward speed. Thus

$$P_T = kV_iT + \frac{1}{8}\sigma C_{D_0}\rho A(\Omega R)^3\left[1 + K(\frac{V}{\Omega R})^2\right] + \frac{1}{2}\rho V^3 f,$$

Where k= 1.15, V_i is the induced velocity, T is the thrust, $C_{D_0} = 0.015$ zero lift drag coefficient, Ω is the angular velocity, R is the radius of the main rotor, K= 4.65, V is the forward speed of the helicopter, ρ is the density at sea level, and $f = 1.15$ is flat plate area.

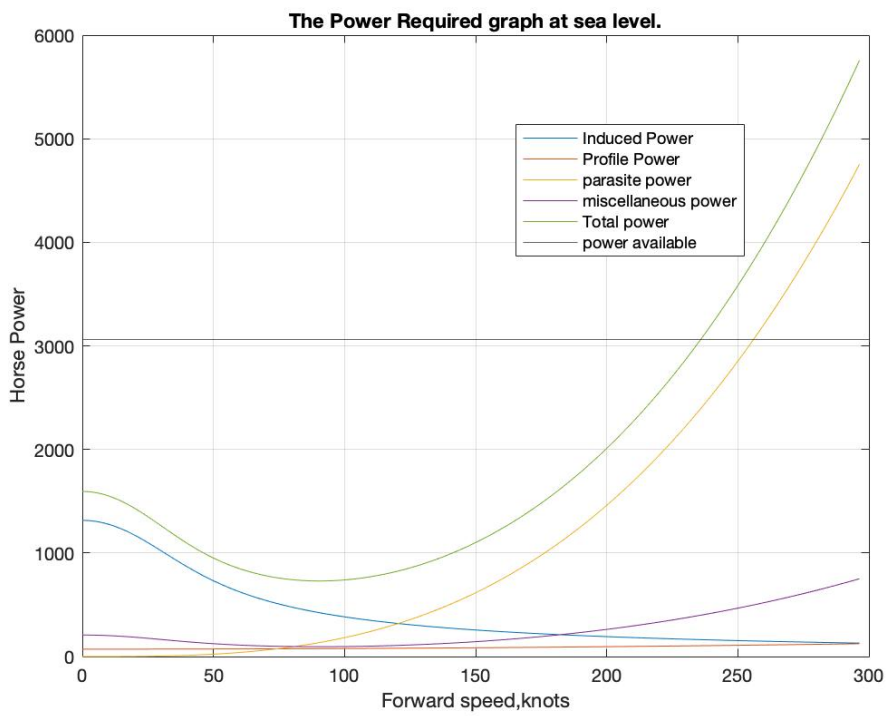


Figure 2 Power vs Forward speed at Sea level

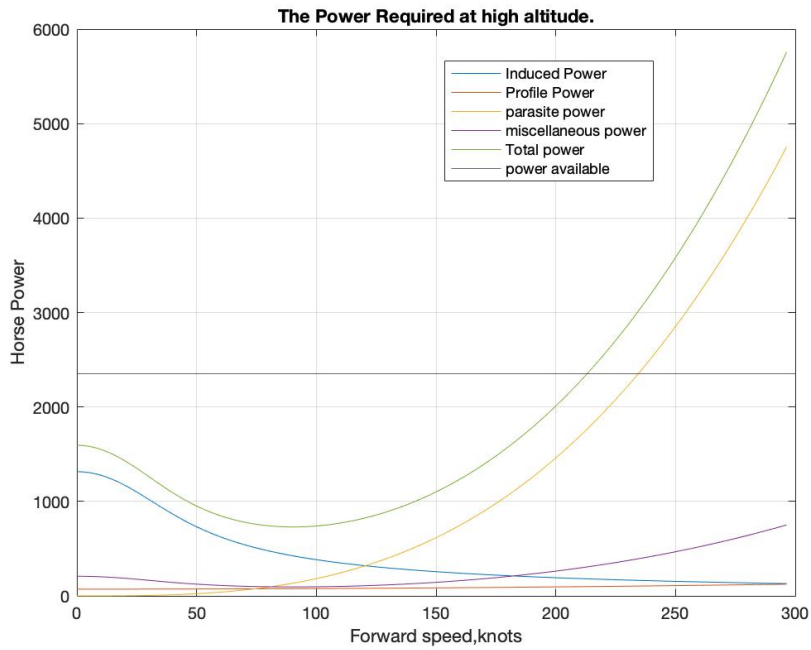


Figure 3 Power vs Forward speed at high altitude

3.4 PERFORMANCE ANALYSIS

Power analysis at sea level is shown in graph 1,

- Velocity for maximum endurance (V_{Emax})= 80 knots
- Velocity for maximum range(R_{Rmax})=115 knots
- Velocity for maximum speed (V^*_{max})= 230 knots

For High altitude, From Graph 2

- Velocity for maximum endurance (V_{Emax})= 70 knots
- Velocity for maximum range(R_{Rmax})=120 knots
- Velocity for maximum speed (V^*_{max})= 210knots

Chapter 4. Additional Design Aspects

4.1 Rotor configuration

For Rotor hub configurations it is important to outline what levels of maneuverability are needed for the required SAR helicopter. Considering search and rescue requires an agile vehicle that can reach difficult areas to traverse and hover over, a conventional main rotor configuration is best suited for our needs.

This type of configuration is not only low cost and simple to design and manufacture, but their designs also provide inherent stability for the rotor hence making flight easy and responsive to input. Their simplicity also allows for low weight distribution and therefore conventional aircrafts can generate a high lift to drag ratio, making flight efficient at both high and low speeds.

Similarly, a rigid rotor hub can provide great mechanical stability and quick responses to inputs given by the pilot. Also comes in lightweight variations that further reduces gross weight and minimizes drag.

Tail rotor configurations are just as important, if not more due to their necessary anti torque function to balance the unwanted torque creation by the main rotor. For similar reasons as the main rotor, we used a conventional type for our tail rotor as well, due to them not only being very lightweight and responsive but also with the added benefit of providing great yaw control for the helicopter

4.2 Main and tail rotor sizing

Through our initial calculations for weight estimations, we used a disk loading of 9.5 to base our helicopter around. Using those calculations, we found our main rotor radius to come to 18 feet. A main rotor radius of 18 feet allowed us to design a helicopter that could lift the helicopter comfortable and achieve an empty weight fraction of less than 0.45.

For the tail rotor radius, it was essential for it to be large enough to produce the required counter thrust to mitigate the fuselage torque, and hence by using estimations we found our radius to be 3.6 feet in length.

4.2.1 Number of Blades

The Number of blades determine the solidity of our helicopter. Various solidities serve different purposes of each helicopters design. A lower solidity would provide less stress on the hub and rotor increasing their lifespan as well as reducing overall noise, however this would result in less generation of lift, limiting our SAR to carry heavy loads. Hence a larger number of blades, mainly 4 blades were selected for our helicopter design as they would not only improve lifting capability in most situations but also drastically improve hover efficiency and increased

maneuverability. For the tail rotor, it was found that 4 blades were enough to minimize drag and tip vortices and generate more than enough counter torque.

4.2.2 Airfoil Selection

Airfoil	**Relative thickness**	$C_{L\,max}$ **at** $M = 0.4$	M_{dd} **at** $C_L = 0$	C_L/C_D **at** $M = 0.6, C_L = 0.7$	C_{m0} **at** $M = 0.4, C_L = 0$
SC1095	8.9	1.2	0.814	60	-0.018
SC1094 R8	9.5	1.3	0.798	55	-0.020
SC2110	9.9	1.5	0.825	70	-0.010
NACA 0012	12	1.3	0.776	47	0
ILH312	12	1.5	0.822	64	-0.060
NPL 9615	11.3	1.0	0.789	57	-0.060

Table 5 Airfoil Selections

The table above shows us the various airfoils we considered for our selection. The airfoil we selected is the NACA 0012 for both the main and tail blades. The NACA 0012 is an extremely reliable and strong contender amongst its competitor airfoils when we observed its performance in SAR helicopters. And after careful consideration we came to conclude why it was the best out of all of them.

The NACA 0012 is a modern symmetric airfoil that has an extremely desirable thickness and lift curve slope.

- The thickness provides a great compromise between lift and drag allowing our design helicopter to be able to hover far more effectively and for longer periods. And also proved to make our helicopter much more fuel efficient.
- Its moderate Lift Curve slope, allows the helicopter to easily achieve hover, takeoff and landing capabilities.
- It can perform extremely well in both low and high speed environments.

- Usage of this airfoil on both main and tail rotors gives our helicopter very good versatility, with its balance between lift and drag both.

4.2.3 Blade twist and design

Blade twist helps further distribute lift along the wingspan of a blade, making high speed forward motion more attainable by delaying blade stall on the retreating end. Blade twist of our blades was chosen to be -10 degrees of linear twist. A twist of this degree reduces power required and further improves the figure of merit during hover. Maximizing forward flight.

4.2.4 RPM and Tip Speed

It is imperative that the design requirements for the maximum main rotor tip speed do not exceed 195 m/s (640 ft/s). This value for tip speed is carefully selected in consideration of noise, stall, and compressibility factors. The acceptable range of tip speed for the helicopter design is demonstrated in the figure below. The design tip speed is 630 ft/s and the cruise speed is approximately 120 knots, which are in compliance with the graph presented.

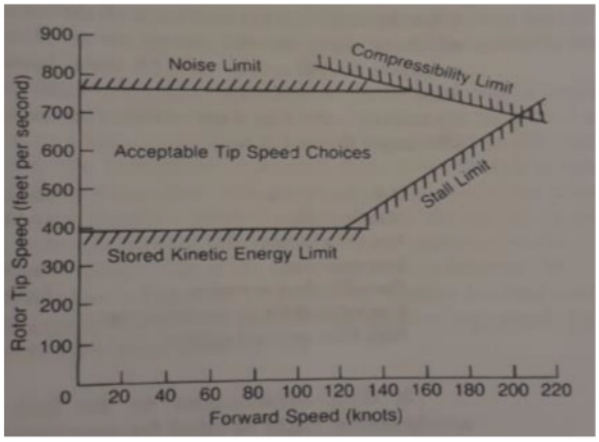


Figure 4 Tip Speed Criteria

Hence, we can calculate our main rotor tip speed as seen below.

$$\Omega = \frac{Vtip}{R} = \frac{630\ ft/s}{18\ ft} \times\ 9.549 = 332.6\ rpm$$

Similarly for the tail rotor and its RPM we need to select a tip speed that does not exceed 590.5 ft/s and since our tail rotor radius is 5 times less than the main rotor diameter, we can expect a RPM of the tail rotor to be much larger. Our tail rotor radius is 3.6 feet long and so

$$\Omega = \frac{Vtip}{R} = \frac{590.5\ ft/s}{3.6\ ft} \times 9.549 = 1566.3\ rpm$$

4.3 Empennage arrangement

The empennage is a group of structural components located at the tail end of an aircraft that enhance stability and control during flight. These features are added to the main fuselage and include horizontal and vertical stabilizers. They aid in maintaining the helicopter's stability, making it easier to control, and improve the overall comfort of the flight. For our design we selected tailplane or conventional type of empennage. A conventional empennage has the benefit of providing very good aerodynamic stability for the vertical and horizontal stabilizers to generate lift and yaw and pitch control. They provide a balanced control surface area for the helicopter to operate with quick responses and manage adverse flight conditions. Not only that but conventional types are also cost effect, simple to manufacture and maintain. They are widely used among all kinds of helicopters due to their reliability.

4.4 Landing gear

Landing gears that provide minimal weight and easy installation and manufacture for SAR helicopters are most often sought for in the market. And the type that serves this the best are the Skid type landing gears. They are strong and give minimal performance drawbacks in most civil helicopters.

4.5 Avionics Suite

An avionics suite refers to a collection of electronic systems used on an aircraft to perform various functions. These systems are typically integrated and controlled by a central computer or display unit. In our design, we use the Honeywell Primus Epic avionics suite since is essential for ensuring the stability and safety of helicopters and those on board. The honeywell Primus Epic avionics suite components include:

- Navigation systems, which provide information on the aircraft's position and allow for communication with air traffic control.
- Communication systems, which facilitate communication between the aircraft and ground stations or other aircraft.
- Weather radar, which detects and displays information about precipitation and other adverse weather conditions.
- Flight control systems, which assist the pilot in maintaining control of the aircraft and following a specific course or altitude.
- Engine and systems monitoring systems, which provide early warning of potential issues and ensure the health of the aircraft's systems.
- Terrain avoidance systems, which use radar or other sensors to detect and display information about terrain, obstacles, and other hazards to flight.
- Emergency locator transmitters (ELT), which automatically transmit a signal to aid in locating the aircraft in case of an emergency.
- Display systems, such as multifunctional displays (MFD), which provide the pilot with a range of information and help in monitoring the aircraft's systems

figure 5 Avionics Suite

Chapter 5. Helicopter Layout

5.1 Cabin Layout

Our Cabin floor was designed to have 9 passenger seats available positioned on one side of the helicopter. While the other side is dedicated to storing the necessary medical equipment and stretchers and to have them be spread out enough to create room for an extra passenger if needed. The seat distance from each row is 2.7 feet away and all seats are compact and designed to save room and spread weight evenly around the helicopter. Similar design has been made for the cockpit as well but leaving extra room for the avionics suite to adjust and fit without proving to be uncomfortable for the two pilots operating. Other design structures as well as the cabin layout can be observed in the figure below.

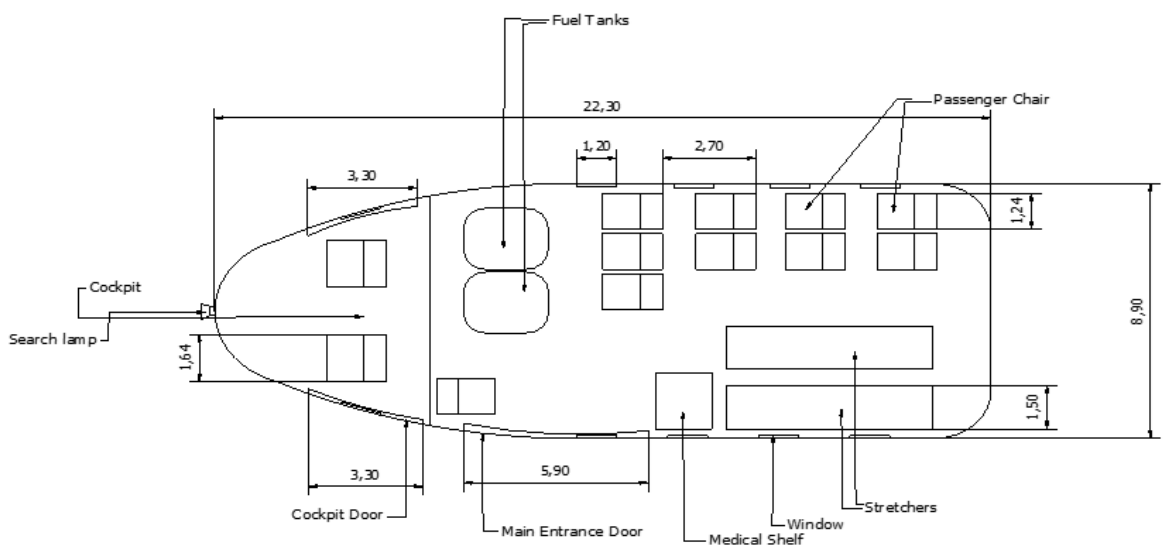
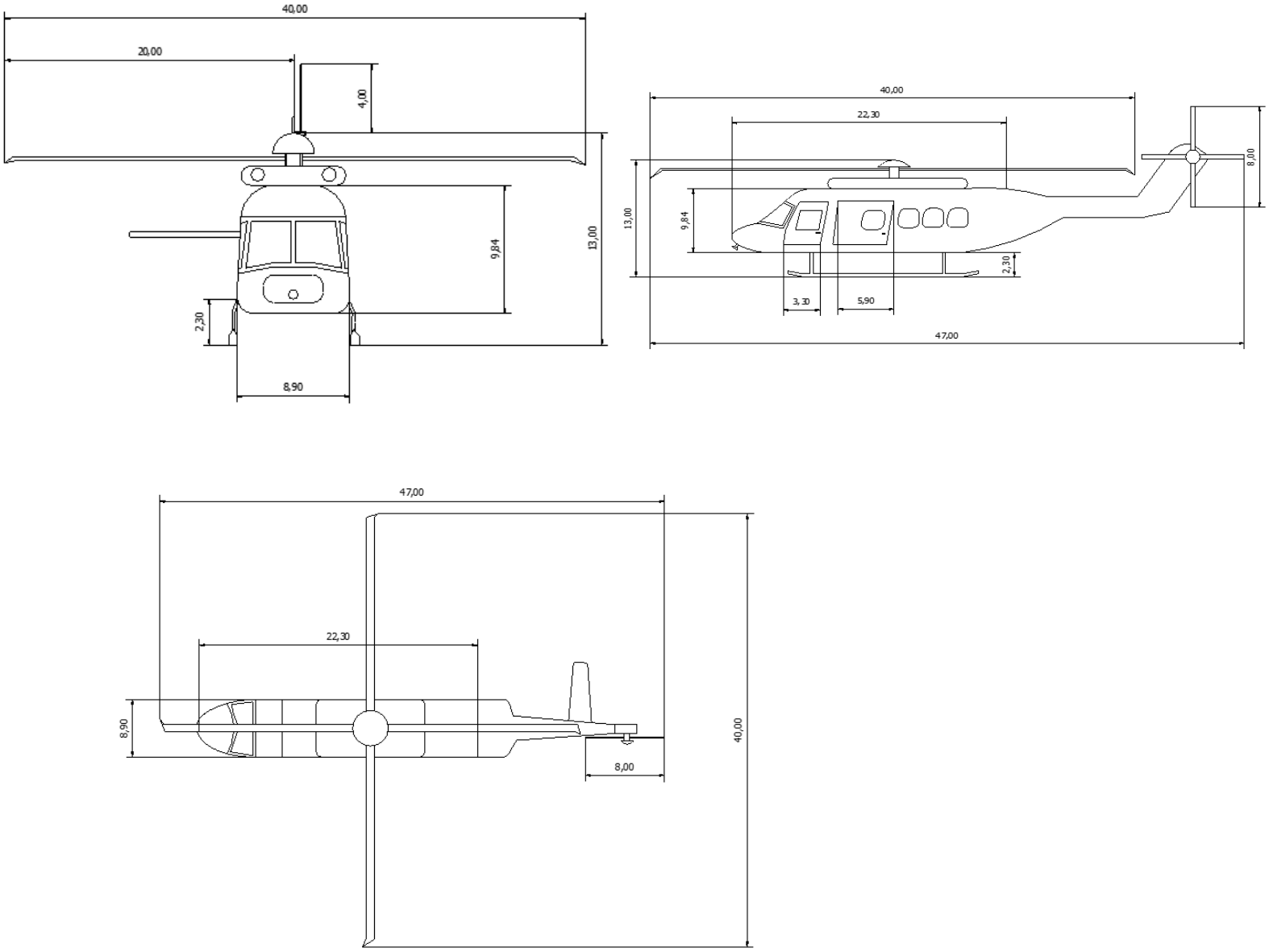


Figure 6 Seating and Layout arrangement

5.2 3D Views



Figures 7 front, side, and top views

5.3 Design Parameters

Pilots	**2**
Passengers	**10**
Range (Km)	**430**
Disk Loading (lb/ft2)	**9.5**
Empty Weight (lb)	**5543**
Max TOGW (lb)	**12355**
Cruise Speed (knots)	**115**
Maximum Speed (knots)	**230**

Table 6 General Parameters

No. of Engines	**2**
Power Plant	**Pratt & Whitney Canada PT6C-67C**
Power (hp)	**1531**
Fuel Capacity (liters)	**14500**

Table 7 Engine Parameters

Length (ft)	**47.0**
Width (ft)	**8.9**
Height (ft)	**13.0**
Main Rotor Diameter (ft)	**36.0**
Tail Rotor Diameter (ft)	**7.2**

Table 8 Geometric Parameters

Conclusions

This project aimed to design a search and rescue helicopter that can operate in Oslo, Norway. Our team first examined competing helicopters to guide the direction of the project. We established design requirements and selected the engine. Secondly, we chose the NACA 0012 airfoil for their design due to its high maximum lift coefficient, which allows for a lower solidity and lighter aircraft. This airfoil is also more efficient, resulting in less fuel consumption and lower power requirements. Additionally, the design features a negative twist, giving the helicopter a good figure of merit.

Then we calculated the gross take-off weight of the helicopter by summing up various weight components, including empty weight, fuel weight, payload weight, and crew weight. We estimated the empty weight by combining eight different weight groups. Calculations were carried out for three different disc loading options, and the option that resulted in a lighter helicopter was chosen. The team also determined the tail and main rotor radius, as well as power for the helicopter in hover

They also considered additional design elements and explained the choices for features such as the rotorcraft configuration, rotor size, tail design, and avionics. The final design was presented with a layout showing the chosen dimensions and decisions. The objective was to create the most suitable, modern, economical, safe, and attractive design while keeping cost in mind. The final recommendation was to use cost-effective materials in the manufacturing process to make the helicopter more affordable to purchase, operate, and maintain. The results were comparable to those of the competitor helicopters, further validating the design.

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Appendix

```
clc
clear all
close all
format short

%%parameters

W_payload=2446 % in lb
W_fuel=2767 % in lb
W_baggage=1600 % in lb
MT=25000 %esttimation take off gross weight

Disc_loading=9.5


R=sqrt(pi*Disc_loading);% radius in meter

A=pi*(R)^2
C=0.95
Nob=4
S=(Nob*C*R)/A
Rt=R/5
Ctail=0.95
Atail=pi*(Rt)^2
Stail=(Nob*Ctail*Rt)/Atail
Thrust=MT
RPM=600
Omega=(2*pi*RPM)/60
Vtip=Omega*R
HP=3062

%dimensions
Lenght=47
Height=12
Width=8.9

Vc=225
PLT=Thrust/HP
Wg=10000;
Wg_final=15000;
while abs(Wg-Wg_final)>1
```

```
W_r = 1.7*(Wg^0.342)*(R^1.58)*(S^0.63); % 1 main rotor
group
W_tr = 7.12*((Wg/1000)^0.446)*(Rt^1.62)*(Stail^0.660); % 2
Tail rotor group
Wgo=Wg/1000;
W_t = (166*Wgo)/(PLT);  %  Power plant group

W_ps = 0.00155*(Wg^1.07)*(Disc_loading^0.54); %  power
plant section group
W_ds = 42.4*((HP*R)/Vtip)^0.763; %  Mechanical drive system
group
W_fc = 0.0226*(Wg^0.712)*(Vc^0.653); %  Flight control
group
W_lg = 0.0475*(Wg^0.975); % Landing gear group
W_f =
0.0382*(Wg^0.598)*(Lenght^0.942)*(Height^0.295)*(Width^0.45
3); % 8 Fuselage group
W_p = 0.146*(Wg/PLT); % Forward propulsion group
W_empty = W_r+W_tr+W_t+W_ps+W_ds+W_fc+W_lg+W_f+W_p;
W_empty = 1.1*W_empty; % other equipment = 8% of total
weight
Wg_final = W_empty + W_payload + W_fuel+W_baggage;
Wg = Wg+1;
end
Wg = Wg-1
empty_weight = W_empty/1.08
empty_weight_fraction=W_empty/Wg_final
T= 12739; % Thrust is equal to the weight in Hover in lb

R = 20; %radious of blade
c = 0.95;  %chord of blade
omega = 600*2*pi/60; %rotational speed of blade
Cdo =0.001 ; %drag coefficient
k = 1.15 ;
K= 4.65;
Nb = 4; % number of blades
rho = 0.00208; % density in slugs/ft^3
A = pi*R*R; %area of blade
sigma= (c *Nb)/(pi*R); % solidity
f= 0.016*A;
Pavailable= 3062 ; % power available
% induced power
V = [0:1:500]; %velocity in ft/sec
```

```
Vi= sqrt((-0.5.*V.^2) + 0.5*sqrt((V.^4)+
4*(T/(2*rho*A))^2));
Pinduced= k.*Vi*T;
PinducedHP= Pinduced*1/550; % induced power in Horsepower
% profile power
Pprofile = ((0.125 *sigma *Cdo *rho *A *(omega*R)^3)*(1
+(K.*(V./(omega *R)).^2)));
PprofileHP = Pprofile/550; % profile power in Horsepower
% Parasite power
Pparasite=0.5 *rho .*V.^3*f;
PparasiteHP= Pparasite*1/550 ; % parasite power in
horsepower
% miscellaneous power
Pmiscel =0.15*(Pinduced+Pprofile+Pparasite);
PmiscelHP =Pmiscel*1/550; % miscellaneous power in
horsepower %total power
Ptotal = Pinduced+Pprofile+Pparasite+Pmiscel;
PtotalHP = Ptotal*1/550; % total power in Horsepower
Vknot= V.*1/1.687; % velocity in knots
plot(Vknot,PinducedHP)
hold on
plot(Vknot,PprofileHP)
hold on
plot(Vknot,PparasiteHP)
hold on
plot(Vknot,PmiscelHP)
hold on
plot(Vknot,PtotalHP)
hold on
yline(Pavailable)
hold on
grid on
title('The Power Required graph at sea level.')
xlabel('Forward speed,knots')
ylabel('Horse Power')
legend('Induced Power','Profile Power ','parasite
power','miscellaneous power','Total power','power
available')
```